

National Institute of Technology, Kurukshetra-136119

Mid Term- I, Even Semester 2025-2026

Subject: Decision Making Using AI (AIPC 200)

Time: 50 Minutes

Maximum Marks: 15

Branch: B.Tech. AIML (IV Semester)

1	In 8-Queens using hill-climbing, if probability of success in one run is $p = 0.14$, compute the expected number of random restarts required to find a solution.	3
2	If h_1 and h_2 are admissible, which of the following are also guaranteed to be admissible and which are not? Justify: (i) $h_1 + h_2$, (ii) $h_1 * h_2$, (iii) $\max(h_1, h_2)$, (iv) $\min(h_1, h_2)$	4
3	Which following graph search methods which are guaranteed to output optimal solutions to the problem and which are not (justify your answer): (i) UCS, (ii) DFS, (iii) A* (with a consistent and admissible heuristic), (iv) A* (with heuristic that returns zero for each state)	4
4	In the following figure S is the start state and T is the goal state. The edges are labeled with distances, and the nodes are labeled with a heuristic estimate to your destination at T. Compute the following to estimate cost to reach to goal and show the path as well: i) A* algorithm with the heuristic values indicated on the graph. Show the nodes in the list when the goal is reached. ii) Draw the search tree using DFS and show final path from S to T. (Always expand the node with the alphabetically smaller label first)	4

